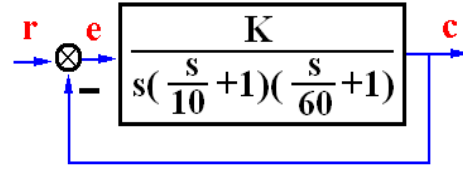


频率法串联迟后超前校正

例 1 已知单位反馈系统的开环传递函数：

$$G(s) = \frac{K}{s(\frac{s}{10} + 1)(\frac{s}{60} + 1)}$$

要求：
$$\begin{cases} r(t) = t \rightarrow e_{ss}^* \leq 1/126 \\ \gamma^* \geq 35^\circ \\ \omega_c^* \geq 20 \text{ rad/s} \end{cases}$$



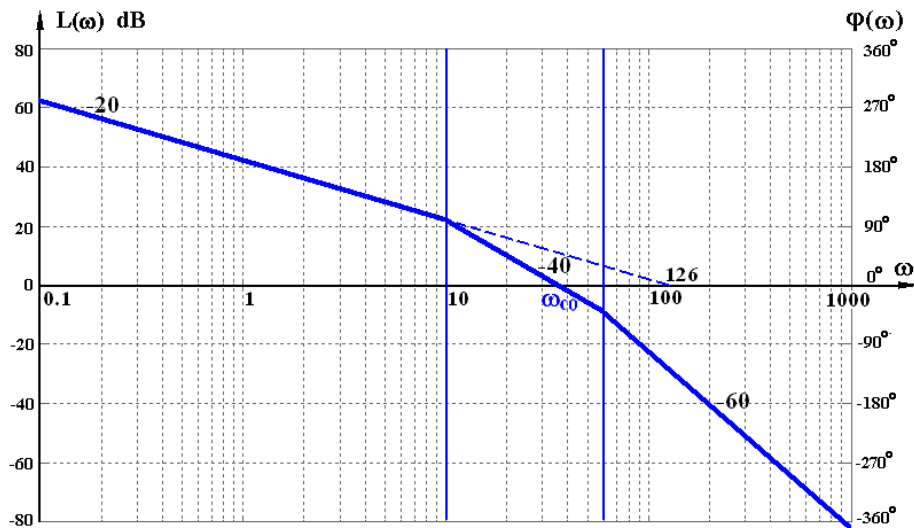
确定校正装置的传递函数 $G_c(s)$ 。

解：
$$G(s) = \frac{K}{s(\frac{s}{10} + 1)(\frac{s}{60} + 1)}$$

$$\begin{cases} \text{开环增益} = K \\ \text{系统型别} = 1 \end{cases}$$

① $e_{ss}^* = \frac{1}{K} \leq \frac{1}{126}$ ，取 $K = 126$ ；

② 作 $L(\omega)$ 如图所示：



$$\omega_{c0} = \sqrt{10 \times 126} = 35.5$$

$$\begin{aligned} \gamma_0 &= 180^\circ - 90^\circ - \arctan \frac{35.5}{10} - \arctan \frac{35.5}{60} \\ &= 180^\circ - 90^\circ - 74.3^\circ - 30.6^\circ = -14.9^\circ \end{aligned} \quad (\text{系统不稳定})$$

③ 确定校正形式：

● 用超前: $\varphi_m = \gamma^* - \gamma_0 + 10^\circ = 35^\circ + 14.9^\circ + 10^\circ = 59.9^\circ$ (不行)

● 用迟后: 在 $\omega_c^* = 20$ 处

$$\begin{aligned}\gamma_0(20) &= 180^\circ - 90^\circ - \arctan \frac{20}{10} - \arctan \frac{20}{60} \\ &= 90^\circ - 63.4^\circ - 18.4^\circ = 8.2^\circ < 35^\circ = \gamma^*\end{aligned}\quad (\text{也不行})$$

决定采用迟后—超前校正, 取 $\omega_c = \omega_c^* = 20$ 进行设计。

④ 确定 $G_c(s)$:

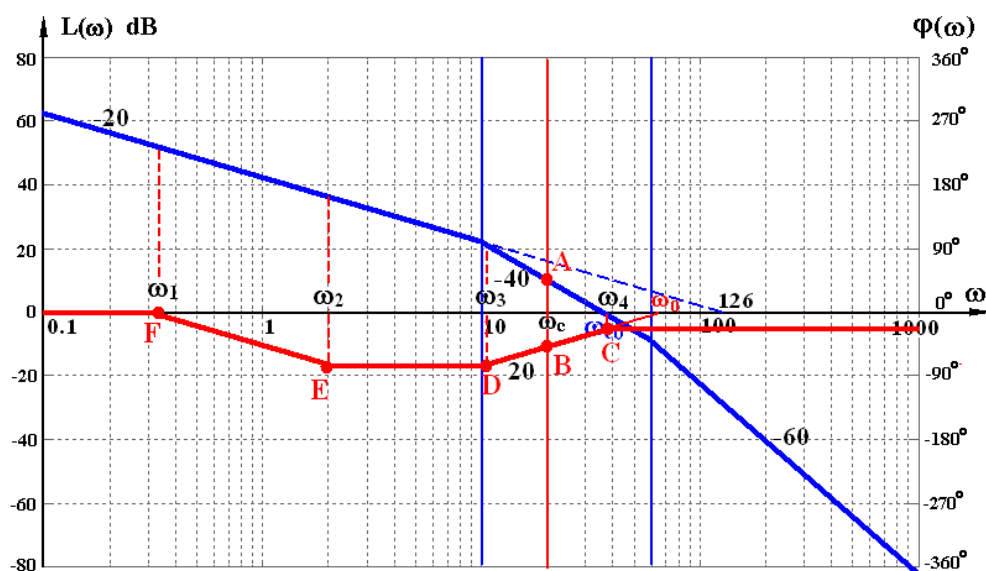
计算超前部分应提供的超前角:

$$\varphi_m = \gamma^* - \gamma_0(\omega_c) + 6^\circ = 35^\circ - 8.2^\circ + 6^\circ = 32.8^\circ$$

$$a = \frac{1 + \sin \varphi_m}{1 - \sin \varphi_m} = 3.4 \quad \rightarrow \quad \sqrt{a} = \sqrt{3.4} = 1.85$$

$$10 \lg a = 10 \lg 3.4 = 5.32 \text{ dB}$$

作图: 在 $\omega_c = 20$ 处设计 $\rightarrow$ A, B



在 B 点左右拉 $\sqrt{a}$ 倍频 (或上下定 5.32 dB) $\rightarrow$ C, D

$$\text{C 点: } \omega_c = \sqrt{a} \omega_c = 1.85 \times 20 = 37$$

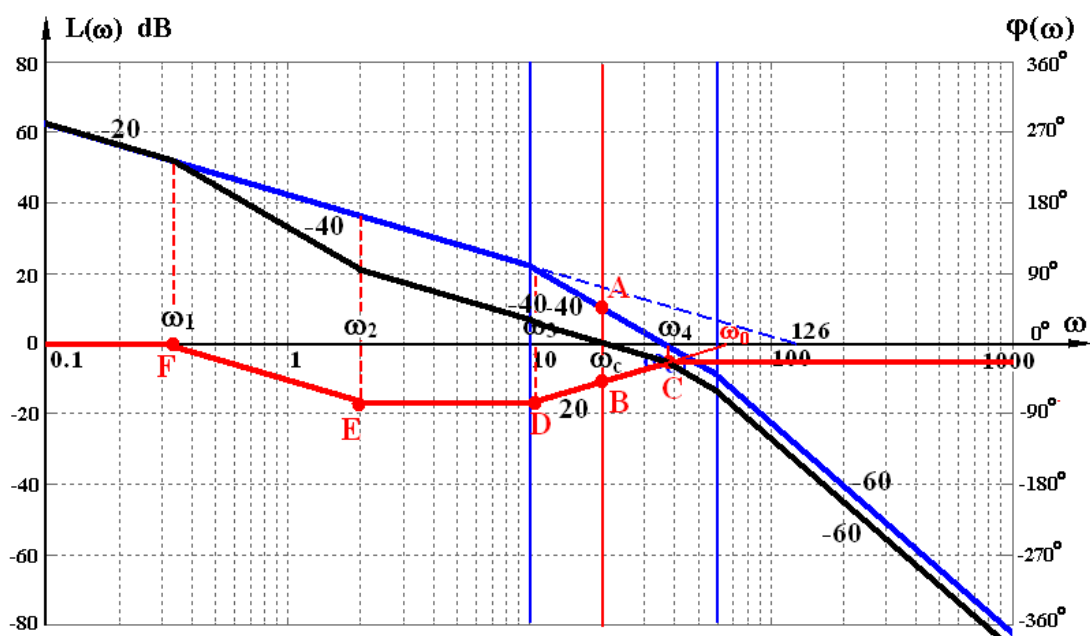
$$\text{D 点: } \omega_D = \omega_c / \sqrt{a} = 20 / 1.85 = 10.81$$

定 E: E 点: $\omega_E = 0.1\omega_c = 0.1 \times 20 = 2$

$$\text{F 点: } \frac{\omega_0}{\omega_{c0}} = \frac{\omega_{c0}}{\omega_c} \quad \omega_0 = \frac{\omega_{c0}^2}{\omega_c} = \frac{35.5^2}{20} = 63$$

$$\frac{\omega_0}{\omega_D} = \frac{\omega_E}{\omega_F} \quad \omega_F = \frac{\omega_D \omega_E}{\omega_0} = \frac{10.81 \times 2}{63} = 0.343$$

$$G_c(s) = \frac{(\frac{s}{\omega_E} + 1)(\frac{s}{\omega_D} + 1)}{(\frac{s}{\omega_F} + 1)(\frac{s}{\omega_C} + 1)} = \frac{(\frac{s}{2} + 1)(\frac{s}{10.18} + 1)}{(\frac{s}{0.343} + 1)(\frac{s}{37} + 1)}$$



⑤ 验算:

$$G(s) = G_c(s) \cdot G_0(s) = \frac{126 \cdot (\frac{s}{2} + 1)(\frac{s}{10.18} + 1)}{s(\frac{s}{10} + 1)(\frac{s}{60} + 1)(\frac{s}{0.343} + 1)(\frac{s}{37} + 1)}$$

$$\omega_c = \omega_c^* = 20$$

$$\begin{aligned}
 \gamma &= 180^\circ + \arctan \frac{20}{2} + \arctan \frac{20}{10.18} \\
 &\quad - 90^\circ - \arctan \frac{20}{10} - \arctan \frac{20}{60} - \arctan \frac{20}{0.343} - \arctan \frac{20}{37} \\
 &= 180^\circ + 84.3^\circ + 61.6^\circ - 90^\circ - 63.4^\circ - 18.4^\circ - 89^\circ - 28.4^\circ \\
 &= 36.6^\circ > 35^\circ = \gamma^*
 \end{aligned}$$

